

WEEK 1

BMI Brain Machine Interface.

Machine ↔ Brain

~ ^(pacemaker) Marcagatos et al for Parkinson → Deep Brain Stimulation
(ameliorating Parkinson's symptoms)
Symptomatic Repair

~~Brain~~ ~~Machine~~

- 1) Teleintelligency - nanotechnology
- 2) Real-time multi signal processing methods
- 4) Sensitive (closing the loop)

light

OPTOGENETICS

(Frontier 4)

Optical stimulation - insert probes to neurons → transform light into electrical activity
"light activated cells"
embed particular ion channels.

• Genetically manipulate ion channels

SIMULATION OF THE BRAIN - BLUE BRAIN PROJECT (Frontier 5)

- 1) Mathematical models of neurons' spiking activity

CONNECTOMICS

3D reconstruction of the brain (EM → electron microscope)

BRAINBOW TECHNOLOGY

Genetic staining of neurons in vivo